Summary Post
by Abdulla Husain Salem Hadna Almessabi - Tuesday, 12 August 2025, 4:35 AM

Agent-Based Systems and Their Rise in Organisations
With its many advantages, including greater flexibility, scalability, and responsiveness in real time, agent-based systems have become critical in the management of complex, decentralised environments (Proselkov et al., 2024). As indicated in my initial post, ABS was not developed to oppose the traditional, centralised systems due to the limitations that are experienced in the top-down decision-making, which fails to perform well in a dynamic environment. ABS enables the autonomous agents to sense the environment, make decisions themselves, and coordinate their actions with other people towards accomplishing their goals without direct involvement of human beings (Rizzati & Landoni, 2024). This is very important when it comes to other operations such as the financial market, supply chain management, and health sectors that need to be able to make decisions promptly as well as bend toleration (Schimeczek et al., 2023).
ABS can be used to model emergent behaviours and predict macro trends like market volatility. ABS, based on the research of AI, allows decentralised decision-making and fluidity and generates resilient, self-governing digital ecospheres. Such systems promote decision making, innovation, and foresight strategically, which places organisations in advantageous positions to be competitive (Terán et al., 2023). The ability of ABS to scale up and fault tolerance on how the systems can be scaled up without the need of having to redesign all the systems in order to accommodate more agents. ABS enables quick reaction to decisions as the decision-making process can be decentralised, without which there is a bottleneck (Zhang & Han, 2024). Also, cooperation or competition can be simulated, contrary to real-world dynamics, and give an idea about socio-economic behaviours using multi-agent systems.

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